

PHYSIOLOGY OF CELL

THE FUNDAMENTAL UNIT OF LIFE

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graph TD; CELL[CELL] --> FR[Functional Regions]; CELL --> CO[Cell Organelles];
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CELL

Functional
Regions

Cell
Organelles

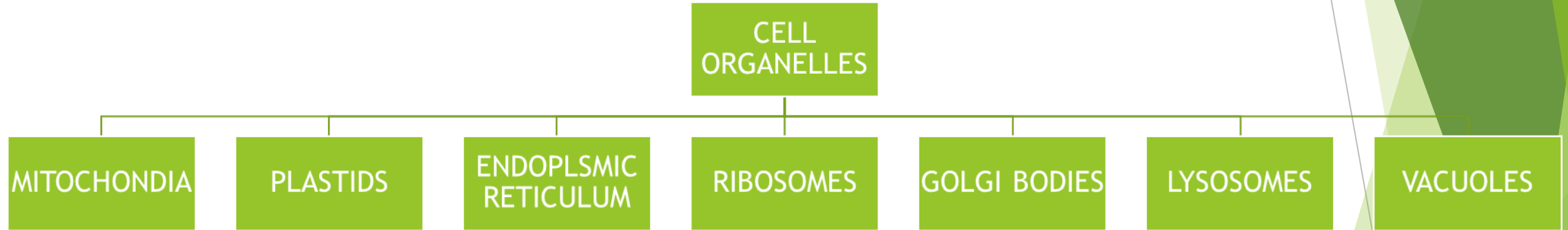
Functional Regions

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graph TD; A[Functional Regions] --> B[Plasma Membrane]; A --> C[Nucleus]; A --> D[Cytoplasm];
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Plasma
Membrane

Nucleus

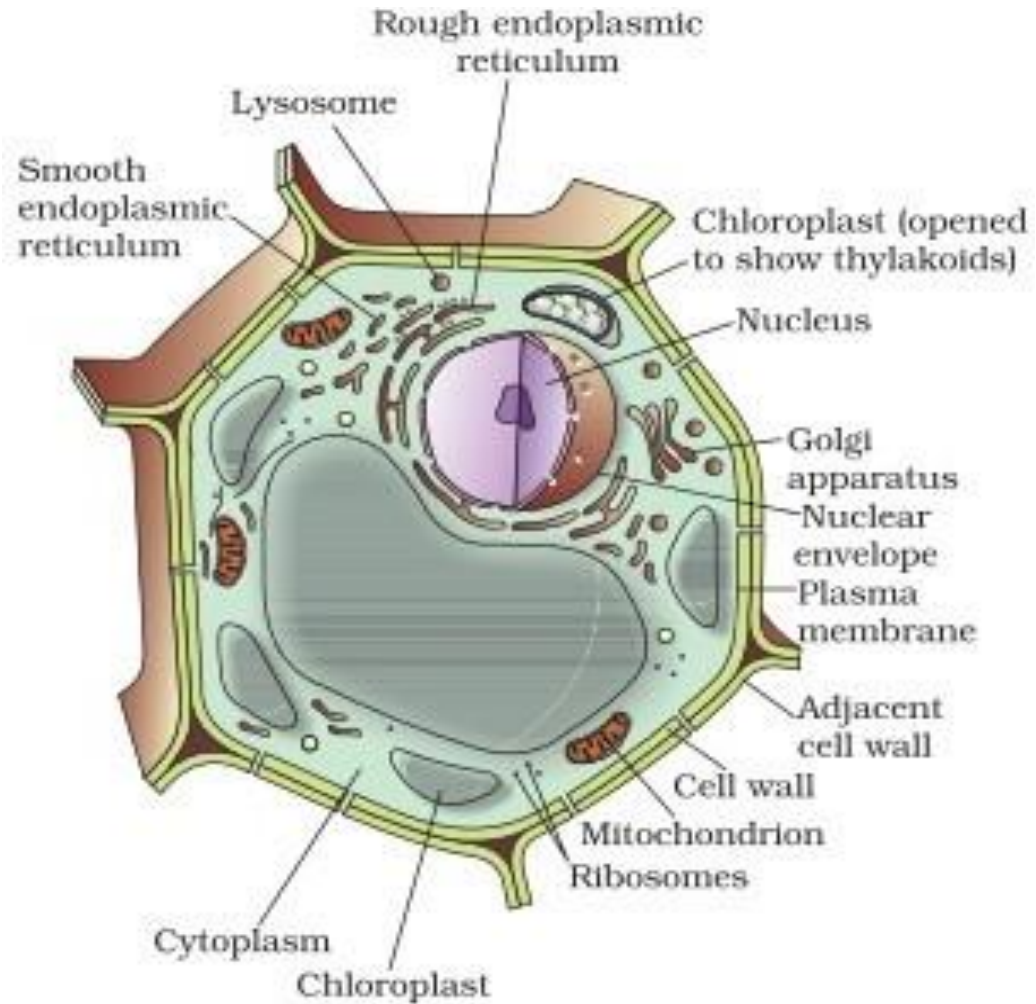
Cytoplasm



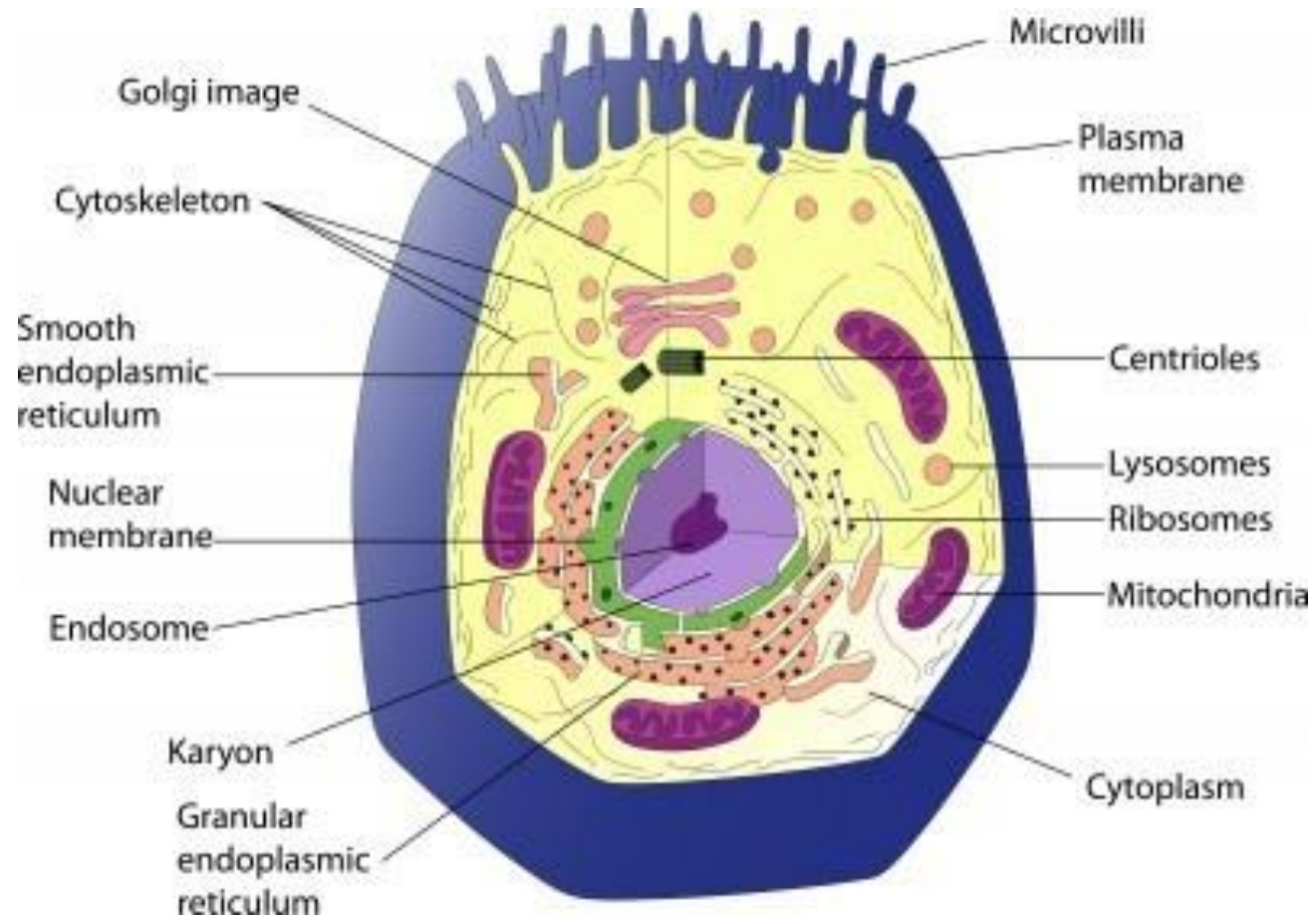
WHAT IS CELL ?

- u It is the structural and fundamental unit of life.
- u Cell have two parts:-
 - 1.Functional regions
 - 2.Cell organelles
- u NOTE:- As cell can exist independently but organelles or functional regions like mitochondria, nucleus, etc. cannot exist independently.

PLANT CELL



ANIMAL CELL



FUNCTIONAL REGION OF CELL

FUNCTIONAL REGIONS:- PLASMA MEMBRANE

- υ Also called as cell membrane.
- υ It is the outer most covering of each cell, which separates its contents from the surrounding medium.
- υ It is made up of lipids(fats) and proteins(building blocks of the body).
- υ **FUNCTIONS:-**
 1. Regulates the movement of the molecules in and out of the cell
 2. It provides definite shape to the cell

CELL WALL

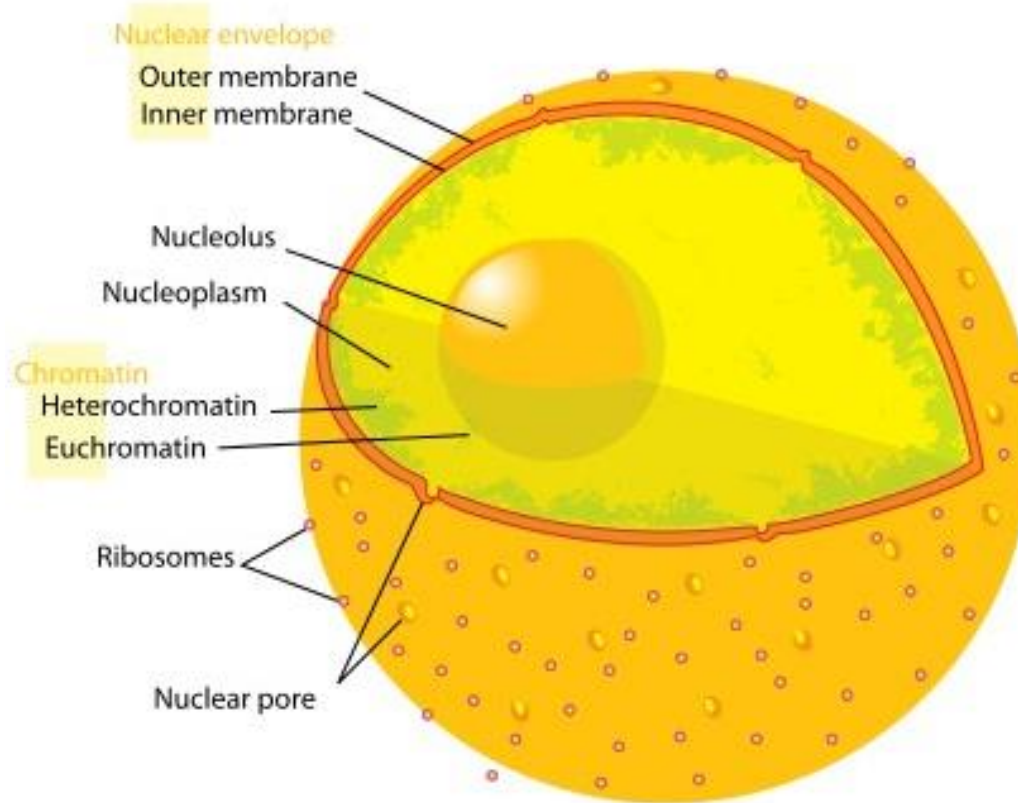
- u This is found in plant cell, fungi cell and bacteria cell.
- u In addition to the plasma membrane, have another rigid outer covering called the plasma membrane.
- u Cell wall mainly composed of cellulose.
- u Cellulose is a complex substance and provides structural strength to the cell.

PLASMA MEMBRANE V/s CELL WALL

PLASMA MEMBRANE	CELL WALL
• PRESENT IN EVERY LIVING CELL • SEMI-PERMEABLE • MADE UP OF LIPIDS AND PROTIENS. • MAIN FUNCTION IS TO REGULATE MOVEMENT OF MOLECULES.	• ADDITION LAYER PRESENT ONLY IN plant cell, fungi cell and bacteria cell. • FREELY PERMEABLE • MADE UP OF CELLULOSE(A CARBOHYDRATE) • MAIN FUNCTION IS TO PROVIDE RIGIDITY(STRENGTH)

NUCLEUS

- Prominent, spherical or oval structure controlling all the cell activities.
- Covered by a double layered membrane called nuclear membrane.



NUCLEAR MEMBRANE

- Nuclear membrane is the membrane containing nuclear pores through which transporting of materials between fluid of nucleus and fluid of cell, takes place.

NUCLEOPLASM

- A colourless dense fluid present inside the nuclear membrane

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graph TD; NUCLEUS[NUCLEUS] --- H[ ]; H --- NUCLEOLUS[NUCLEOLUS]; H --- CHROMATIN[CHROMATIN MATERIAL];
```

NUCLEUS

NUCLEOLUS

CHROMATIN
MATERIAL

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graph LR; A[NUCLEOLUS] --> B["R.N.A.  
(RIBOSE  
NUCLEIC  
ACID)"]; B --> C[PROTIENS  
SYNTHESIS];
```

NUCLEOLUS

R.N.A.
(RIBOSE
NUCLEIC
ACID)

PROTIENS
SYNTHESIS

NUCLEOLUS

- ▮ It is a dense spherical structure(contain R.N.A.) occurring inside nucleus of cells that synthesizes proteins.
- ▮ R.N.A.(**ribose nucleic acid**) : a kind of nucleic acid which is helpful in synthesis of proteins because it contain RIBOSOMES.

CHROMATIN
MATERIAL

CHROMOSOMES

CHROATIDS

CENTROMERE

GENES

D.N.A.

CHROMATIN MATERIAL

υCHROMATIN MATERIAL : An intertwined mass of thread like structures forming chromosomes during cell division.

FUNCTION : Mainly responsible for storing and transmitting the hereditary information from one generation to another.

υCHROMOSOMES : Chromatin condensed into compact rod like bodies at the time of cell division is called as chromosomes.

FUNCTION : Pass hereditary information from one generation to another.

υCHROMATIDS : It is one of the two thread like structures(containing D.N.A.), joined at the centromere, to form a single chromosomes.

υCENTROMERE : It is the point of attachment of two chromatids.

υGENES : Functional unit of chromosomes arranged in single liner order which may be responsible for one or several cell functions

υD.N.A.(**Deoxyribose nucleic acid**) : A kind of nucleic acid acting as a hereditary material and transmitting characters to young ones.

CYTOPLASM

- υ Fluid present outside the nucleus but inside the plasma membrane.
- υ A number of small organelles are found inside the cytoplasm and are know as *cytoplasmic organelles*.

PROTOPLASM

- υ Nucleoplasm + Cytoplasm
- υ Colourless, viscous jelly like semi-fluid, which can be distinguish into a nucleoplasm and cytoplasm

CELL ORGANELLES

CELL ORGANELLES

- ▮ These are living structures of the cytoplasm and are also called as **ORGANOIDS.**

MITOCHONDRIA

- ▮ Mitochondria are the organelles which contain enzymes for cellular respiration in which energy is released.

SHAPE : Rod shaped organelles.

FUNCTION : It provides energy in form of ATP(Adenosine Tri Phosphate) which is vital for the activities of living cells. hence it is also called as "**powerhouse**" or "**powerplant**" of cell.

STRUCTURE : Made up of two membranes, the larger membrane is folded inside the smaller membrane forming finger like projection called as **CRISTAE**.

ENDOPLASMIC RETICULUM

- υ It is a membranous network enclosing a fluid filled lumengiving internal support to the cytoplasm.
- υ **TYPES :**
 - 1) **S.E.R.(Smooth Endoplasmic Reticulum)** : This is smooth because this does not contains Ribosomes on its surface.
 - 2) **R.E.R.(Rough Endoplasmic Reticulum)** : This is rough because it contains Ribosomes on its surface.
- υ **FUNCTION** : Synthesises proteins(by R.E.R.) and Lipids(by S.E.R.) for producing new cellular parts

RIBOSOMES

- Small, bead like bodies found either in free state in cytoplasm or attached to the surface of Endoplasmic Reticulum.

FUNCTION : Associated in the synthesis of proteins.

GOLGI BODIES

- υ Called as **DICTYOSOMES** in plant cells.
- υ Smooth, flattened, sac like structures placed one over other in parallel rows.

FUNCTIONS :

- 1) It packages the synthesised material of Endoplasmic Reticulum and despatches it to various places.
- 2) Involved in the formation of lysosomes and peroxisomes.

LYSOSOMES

- υ Sac like cytoplasmic organelles containing hydrolytic enzymes that degrade those cellular components which become useless.
- υ These serve as intercellular digestive system hence called as **DIGESTIVE BAGS**.
- υ They digest the entire damaged or dead cell contain themselves hence also termed as **SUICIDE BAGS**.

VACUOLES

- υ ONLY FOUND IN PLANT CELL.
- υ IN PLANTS SINGLE PROMINENT VACUOLAE OCCUPIES ABOUT 90% OF THE VOLUME OF THE CELL.
- υ Membranous bags filled with liquid cell sap with single cell membrane called TONOPLAST.
- υ **FUNCTION** : Provides rigidity and turgidity to plant cell.

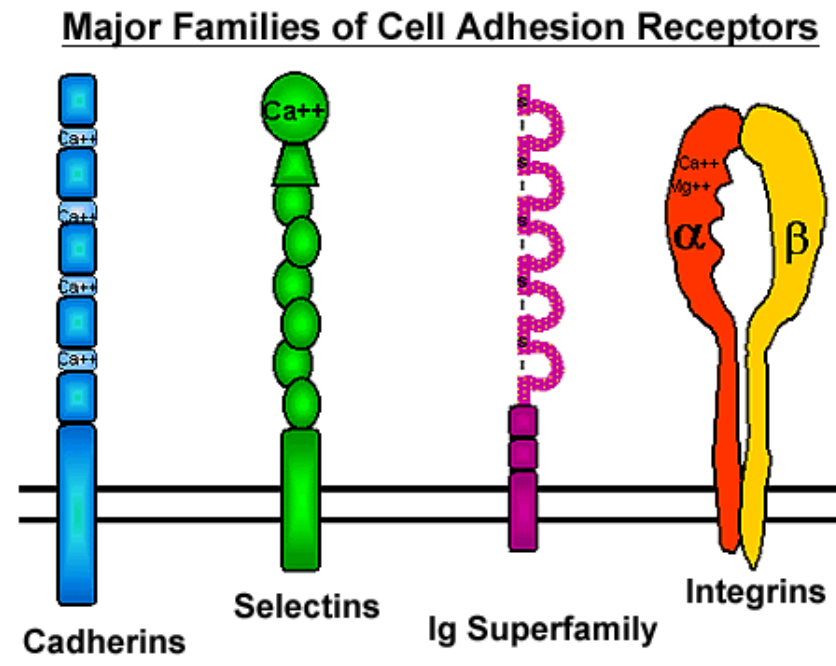
Cell adhesion molecules

Ca²⁺-dependent:

cadherins
selectins
integrins

Ca²⁺-independent:

IgG-like- CAMs (immunoglobulin superfamily)
integrins



Functions of tight junction

Diffusion barrier in the plasma membrane (blocking lateral diffusion in the membrane), the belt-like tight junction divides the plasma membrane into an apical and baso-lateral domain (with different sets of molecules)

- Diffusion barrier in the intercellular space: free diffusion of substances between neighboring cells is blocked.

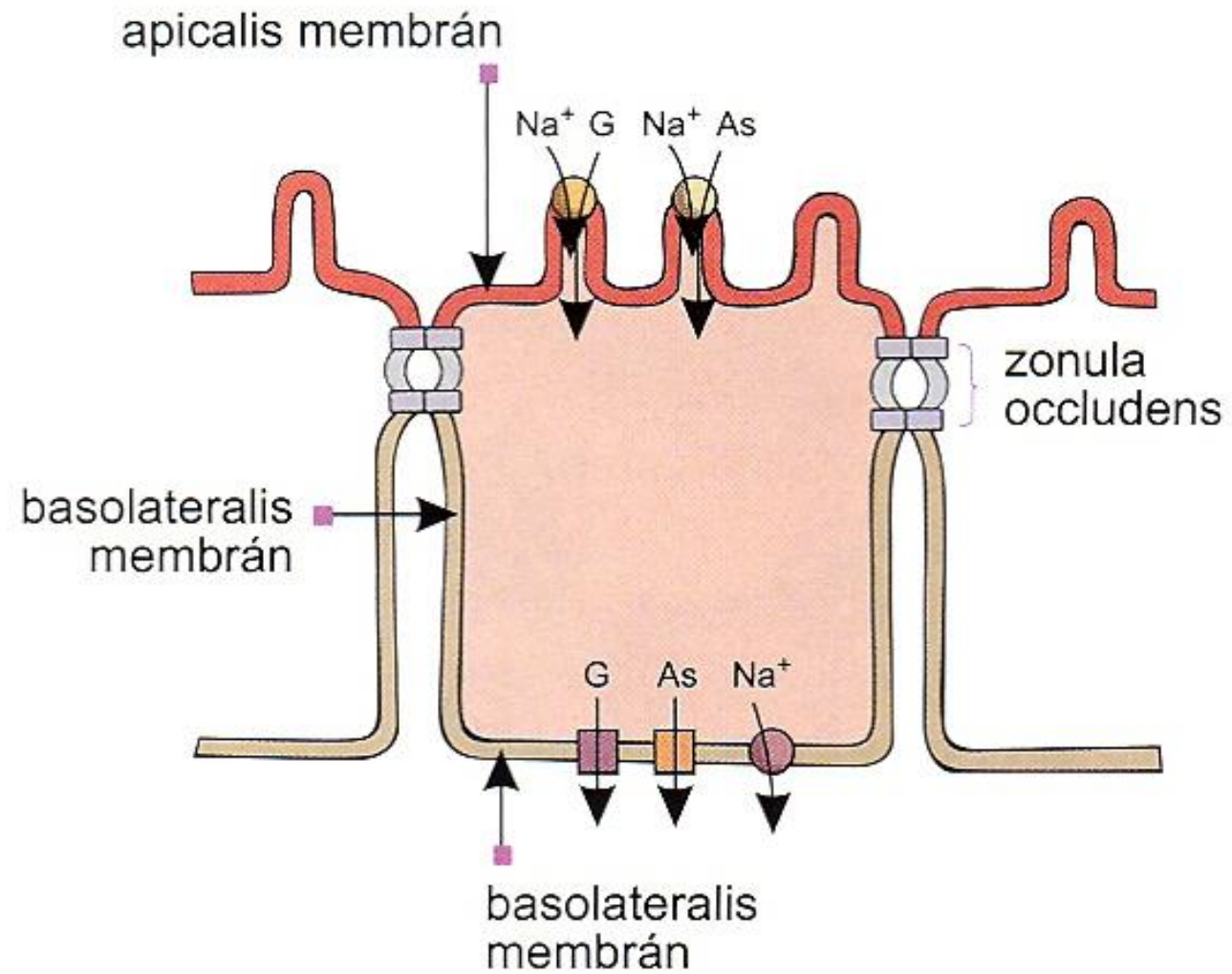
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Biological significance: controlled and unidirectional transport across the epithelial layer

apical membrane

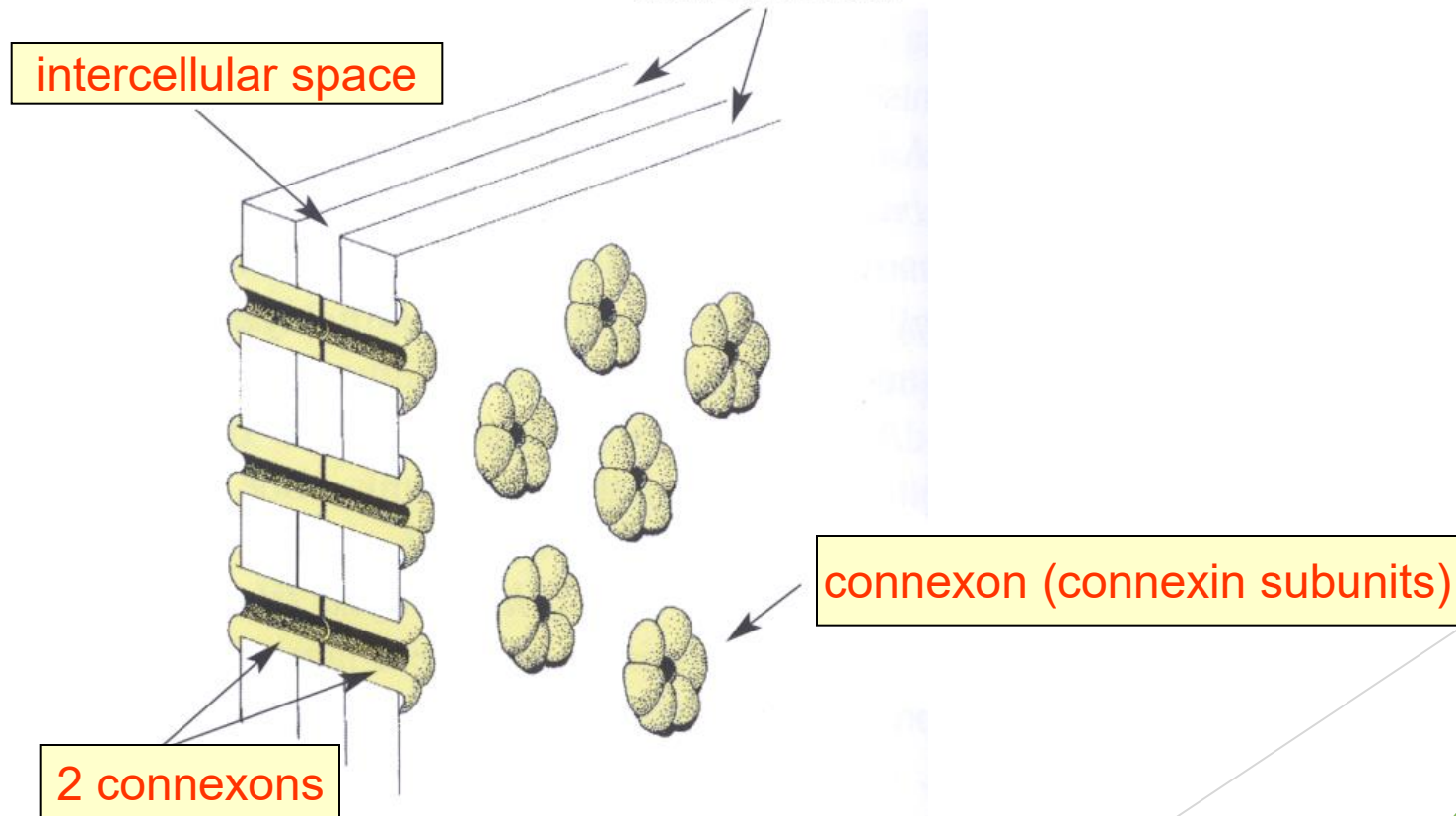
baso-lateral membrane

baso-lateral membrane



Nexus (gap junction)

Patch-like contacts between two cells: large number of *channels (connexons)* in the opposite membranes, that are bound to each other in the intercellular space to form continuous channels between the cells.

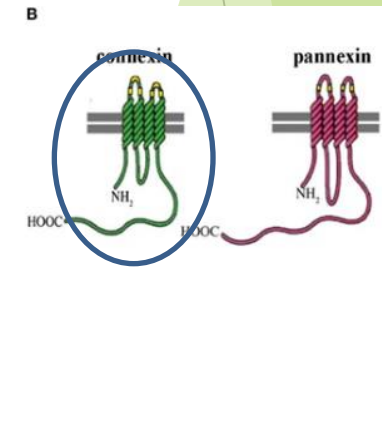
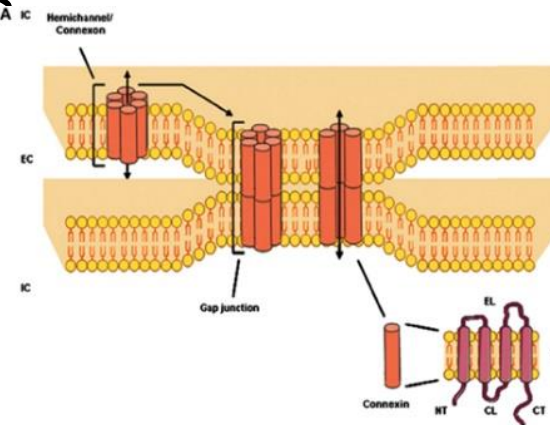


Molecular structure:

connexon: a complex composed of 6 *transmembrane proteins* (connexins), that surround a central canal. A similar complex in the opposite membrane is joined to it in the intercellular space. A *continuous canal* is formed which leads from one cell into the other making communication by free diffusion of low molecular weight substances possible.

Connexin: Transmembrane proteins

Cx32 in nerve fibers,
defects lead to problems in nerve
conduction.



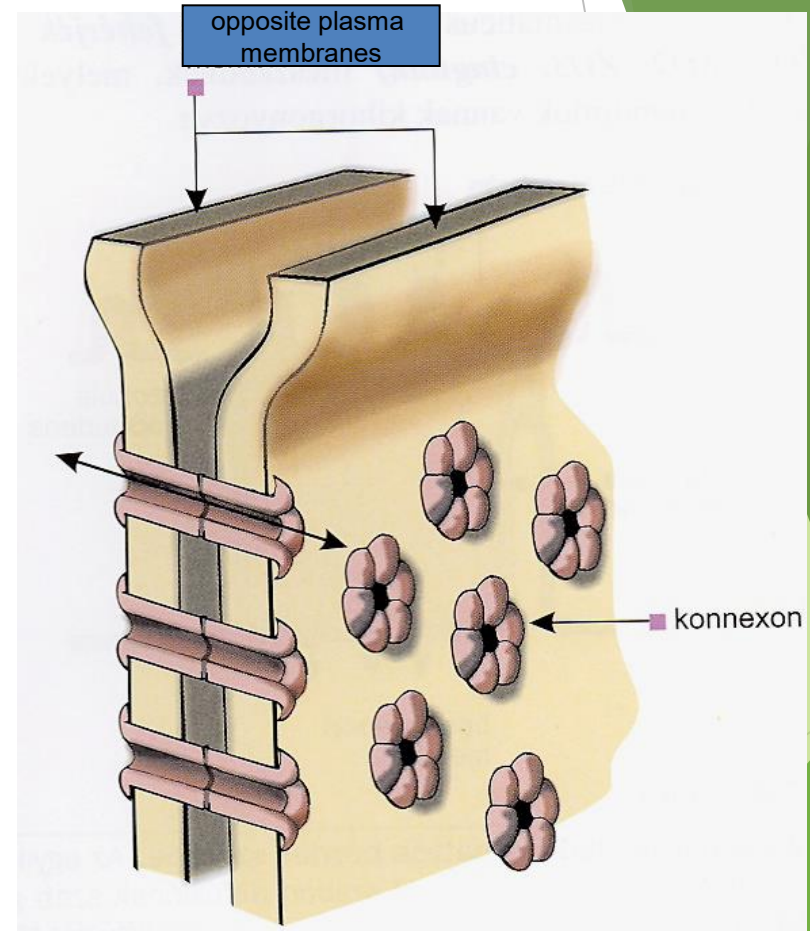
Gap junction: patch-like membrane domain with densely packed connexons

Diversity of connexons, in different cell types. Combinations of different connexins in a connexon.

Examples of connexin isoforms:

Cx50 in the crystalline lens of the eye, defects lead to glaucoma

Cx26 in sensory cells of the inner ear, defects cause auditory malfunction

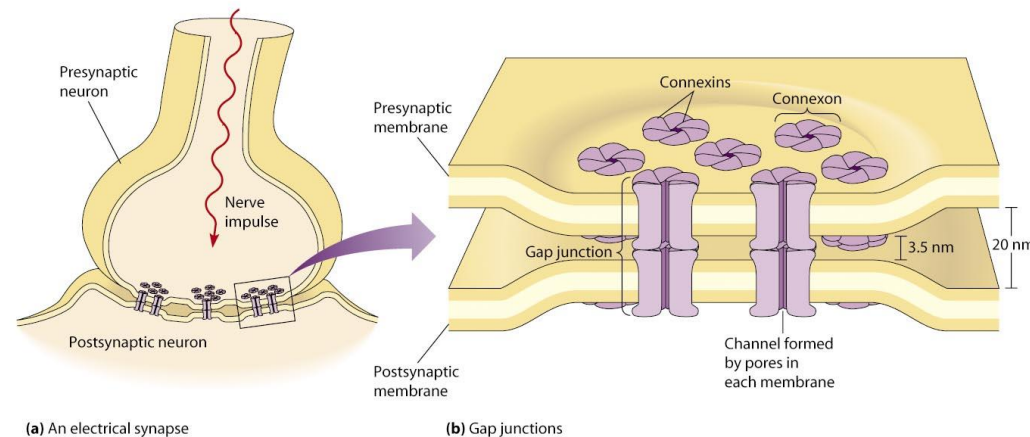


Biological significance:

Communication between cells: nutrient transport (cells of the lens, osteocytes, follicular cells surrounding the oocyte in the ovary), synchronized reaction for signals during the development etc.

Electrical connection between cells: electrical synapsis: ions can migrate between cells, stimulus can be transmitted without delay;

In heart muscle: synchronized contraction



MADE BY :

THANK U

